

Yunwan Fu

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EDUCATION

Shanghai Jiao Tong University

Bachelor of Engineering in Automation (IEEE Pilot Program)

Shanghai, China

Sep 2024 – Jun 2028

Overall Academic Score: 97/100 (**Top 1/25 in Major**)

Relevant Coursework (All A+ Grades): C++ Principles, Video and Multimedia Retrieval, Electronic Circuit Laboratory, Linear and Convex Optimization, Linear Algebra, Probability and Statistics, Mobile Robotics

RESEARCH INTERESTS

Embodied AI, Vision-Language-Action (VLA), Agents, Humanoid, Vision-Language Models (VLMs), and agent harnesses for robotics.

RESEARCH PROJECTS

PSI Lab , USC

Research Intern, Advisor: Prof. Yue Wang

Los Angeles, USA

Mar 2026 – Present

• PhySA: Physical Smoothness Agent for Humanoid Locomotion

- * Designed a dual-channel reward framework decoupling VLM aesthetic scoring from physical smoothness correction, resolving their inherent conflict in humanoid RL.
- * Achieved 59.5% lower joint acceleration, 43% lower mechanical power, and 57% longer episodes over Prompt2Policy on MuJoCo Humanoid-v5.
- * Implemented asynchronous VLM scheduling ($24\times$ fewer calls) and phase-triggered reward editing; manuscript in preparation for CoRL 2026 Workshop.

• Ψ -X: Egocentric Video Annotation via VLM

- * Deployed EgoAnno pipeline with Qwen-30B (vLLM) and DINOv2 on EgoDex dataset; curated long-horizon pick-and-place tasks for targeted evaluation; currently running full pipeline to benchmark zero-shot VLM annotation performance.

• H2: Cross-Embodiment Whole-Body Control Transfer for Unitree Humanoid

- * Migrated NVIDIA SONIC's whole-body policy from Unitree G1 (29 DoF) to H2 (31 DoF) via name-matched weight transfer, retargeted 131,454 SMPL motions into H2 trajectories (100% coverage, 10.9 mm error), and trained four 120k-step policies in Isaac Lab, identifying jitter and sim-to-sim instability as remaining blockers to real-robot deployment.

AWARDS & HONORS

A-Level Academic Scholarship, Shanghai Jiao Tong University

Dec 2025

Rayliant-Sequoia Talent Development Fund

Nov 2025

National First Prize, China Undergraduate Mathematical Contest in Modeling

Nov 2025

“Merit Student” of the 2024-2025 Academic Year, Shanghai Jiao Tong University

Oct 2025

Finalist Award, Mathematical Contest in Modeling (MCM/ICM), COMAP

May 2025

TECHNICAL SKILLS

Languages: Chinese (Native), English (Fluent, CET-6 562/710)

Technical Skills: Python, C++, LaTeX, Linux, ROS/ROS2, Gazebo, OpenCV, MuJoCo, Isaac Sim, PID.